

Improving Long-Context Preference Following in LLMs

Mid-term: Phase 1, reproducing PrefEval baselines on local models

Krishna Jhanwar (12341250)

October 2026

The problem and the plan

Problem (PrefEval, ICLR 2025)

- A user states a preference early (*“I avoid gluten”*)
- Unrelated conversation follows
- Later query: *“Restaurants in Rome?”*
- LLMs often forget the preference as context grows

This talk covers Phase 1: 64,535 questions, 2 models, 5 methods, 2 preference forms, 0.2k–23k tokens, about 71 GPU-hours on one RTX 3060.

Three phases

1. **Reproduce baselines** **done**
2. Train a preference-detection classifier
3. Classifier → preference memory + modern retriever, compared against the Phase 1 baselines

The benchmark task (classification)

- 1,000 preference–query pairs, 20 topics, explicit and implicit preference forms
- Preference at the start → `inter_turns` unrelated LMSYS turns → query with 4 options
- Exactly one option respects the preference; metric = accuracy averaged over 20 topics

Five baselines

- **Zero-shot**: answer directly
- **Reminder**: one sentence asking the model to consider earlier preferences
- **CoT**: five worked examples of preference-aware answers
- **Self-Critic**: answer → critique → revised answer
- **RAG**: top-5 past exchanges retrieved with SimCSE and quoted before the question

Constraint: no Bedrock, one 12 GB GPU

Models from the paper, served locally (Ollama, 8-bit)

- **Llama-3-8B-Instruct**: 8k context → 0.2k–3k rows
- **Mistral-7B-Instruct-v0.2**: 32k context → up to 23k tokens (the proposal's long regime)
- Greedy decoding (temperature 0, seed 41)

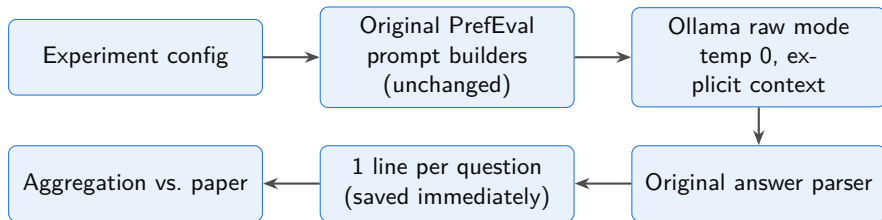
Ruled out

- Mixtral-8x7B / 70B models: 28+ GB, exceeds GPU + RAM

Paper row	Paper turns	inter_turns	Llama-3 tokens	Mistral tokens
0.2k	2	0	275	309
1k	5	3	1,437	1,764
3k	10	8	3,282	4,023
10k	30	28	– (8k limit)	12,139
16k	50	48	– (8k limit)	19,452
23k	70	68	– (8k limit)	26,878

Paper rows mapped to inter_turns, with measured prompt tokens

Pipeline: original prompts, local inference



- Only the inference call is replaced; prompt text is identical to the authors' code
- Resumable after any crash (at most 1 question lost); run manifest with git commit, model digest, settings
- A guard refuses runs whose server settings differ from a model's baselines, which keeps Phase 3 comparisons fair

Issues found and fixed (each would have biased the results)

1. **First attempt failed:** Gemini import, a missing `import random`, three copies of `generate_message`
2. **Double start token:** original prompt + Ollama each add BOS (verified via token counts) → stripped
3. **Empty self-critiques (Llama):** indented template ends in whitespace → model stops at once → trimmed
4. **Silent truncation:** Ollama cuts long prompts *from the start*, where the preference is → explicit context size and checks
5. **Defects in the released data:** one explicit RAG file truncated (19 topics); 7 implicit RAG files end early (973/1000 questions)
6. **32k context did not fit in 12 GB:** flash attention + 8-bit KV cache, used for all Mistral runs

Result: **0** truncated prompts and near-zero parse failures in all runs (except the documented Llama self-critic case).

Explicit preferences: ours vs. paper (Figure 6)

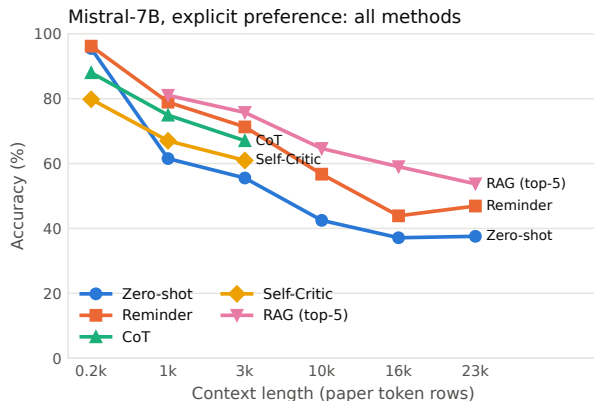


Excluding 0.2k: 12/29 cells within 5 points of the paper, 20/29 within 10 (median gap 6.0). Qualitative trends reproduce.

Where we differ, and why

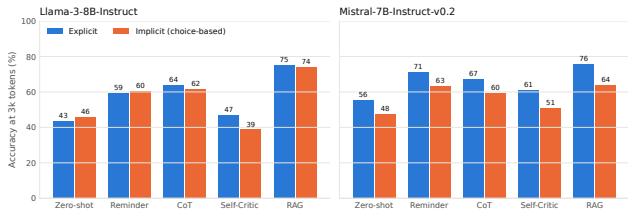
- **Zero-shot, Reminder:** close up to 3k; 8–9 points *higher* at 0.2k (easy setting)
- **CoT:** Llama +11 to +14, Mistral −2 to −6 → differences in model serving, not a pipeline bias
- **Self-Critic:** Mistral within 2–5 points; Llama −15 to −23 at 1k / 3k
 - caused by the original Llama template: 25.2% empty revisions at 1k
 - excluding them: 78.8 / 67.1 / 47.8 (paper 79 / 73 / 62)
- **RAG:** Llama matches; Mistral falls further below the paper as context grows (−6 → −21)
 - plausible cause: 8-bit KV cache at long context (not yet verified)

Long context: Mistral-7B up to 23k tokens



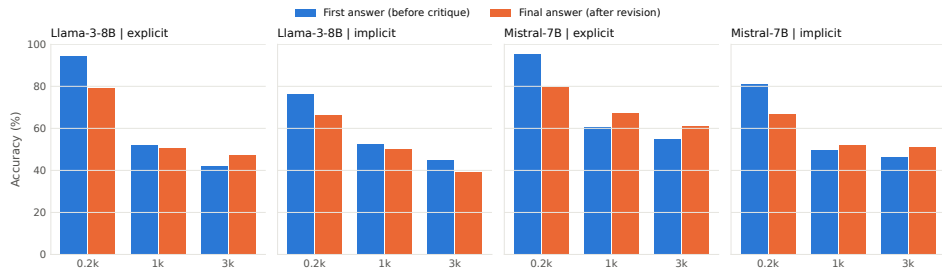
- Steep drop from 0.2k to 10k, then flat
- Reminder > Zero-shot at every length
- RAG best throughout (53.7% vs. 37.6% at 23k)
- **Lesson:** a 3-topic pilot gave $\approx 23\%$ (chance level). Those topics were the hardest; all 20 topics are needed

Implicit (choice-based) preferences: new classification baselines



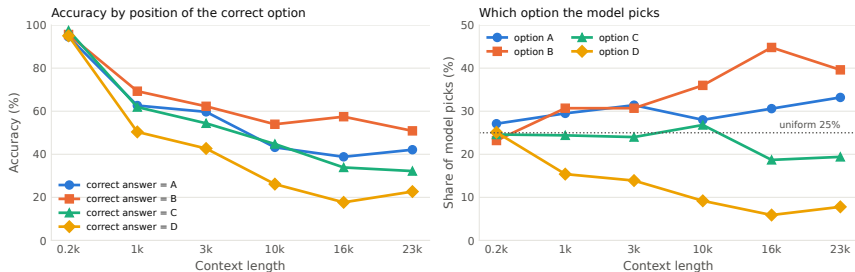
- The paper reports implicit forms only for the generation task, so these are our own baselines
- Mistral: 7–15 points harder everywhere
- Llama: harder mainly at 0.2k; at 3k within 3 points for most methods
- These are the reference numbers for Phase 3

Finding 1: self-critique overturns correct answers



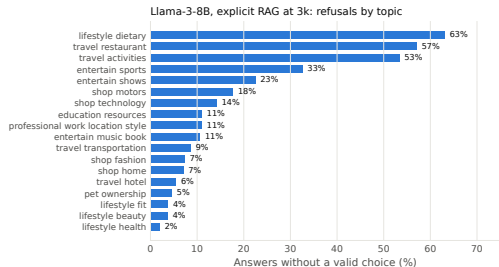
- At 0.2k: Llama 205 right→wrong vs. 51 wrong→right; Mistral 188 vs. 29
- At longer contexts the net effect ranges from -6 to $+7$ points; this matches the paper's observation for classification

Finding 2: at long context the model falls back on answer position

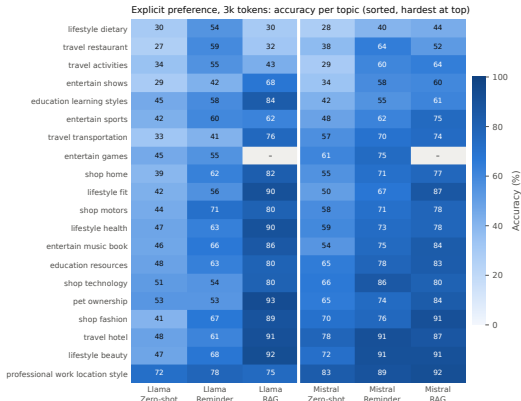


- Accuracy when the correct answer is A vs. D: 60% vs. 43% at 3k; 42% vs. 23% at 23k
- The model favours B and avoids D (picked only 8% of the time at 23k). Once the preference is lost, a position prior remains

Finding 3: retrieved context triggers refusals; topics differ widely



Llama RAG at 3k: 19.1% answers without a choice, mostly refusals (*"I cannot recommend. . ."*); Mistral 0.6%



Deviations and threats to validity

- 8-bit weights on llama.cpp (vs. full precision on Bedrock); Mistral also uses an 8-bit KV cache
- Two prompt normalisations (start token, trailing whitespace); all prompt text is unchanged
- Per-question option shuffle (paired across methods, resumable)
- RAG coverage limited by data defects (19 topics explicit; 973/1000 implicit)
- One deterministic run per cell; binomial standard error ≤ 1.6 points per cell
- Classification task only; the generation task needs an LLM judge

Runs	Questions	GPU hours	s / question
Llama explicit (Tiers A+B)	13,751	9.0	2.3
Llama implicit-choice	13,901	7.6	2.0
Mistral explicit, short (Tiers A+B)	13,853	8.6	2.2
Mistral explicit, long (10k–23k)	8,847	36.3	14.8
Mistral implicit-choice	13,931	9.8	2.5
Smoke tests / checks	195	0.1	2.7
Total	64,478	71.4	4.0

Feasibility and next steps

Phase 2: classifier

- Positives: explicit preference sentences and preference-revealing implicit turns
- Negatives: LMSYS turns and other dialogue turns
- **Topic-level split** to avoid leakage into Phase 3
- DistilBERT / ModernBERT / MiniLM; P/R per form, latency

Phase 3: full pipeline

- Classifier → memory, plus bge / e5 retrieval
- Same runner, prompts and settings guard as the baselines
- Targets: the 0.2k→3k drop, the position-prior collapse, RAG refusals

Scope changes

- Long regime via Mistral; Mixtral dropped (hardware)
- Generation task / proprietary model on a small subset, depending on API budget

Summary

- PrefEval's classification benchmark now runs **fully locally** with the original prompts, and is resumable and logged
- 64,535 questions: 2 paper models $\times$ 5 methods $\times$ 2 preference forms $\times$ up to 6 context lengths
- The paper's findings reproduce; most numbers are within a few points, and the larger gaps have identified or plausible causes
- Found and fixed 3 prompt-formatting issues and 2 defects in the released data
- New observations (self-critique overturning answers, position-prior collapse, RAG refusals) give concrete targets for the classifier + memory pipeline

Thank you. Questions?